



CLDV7111

Assignment 2

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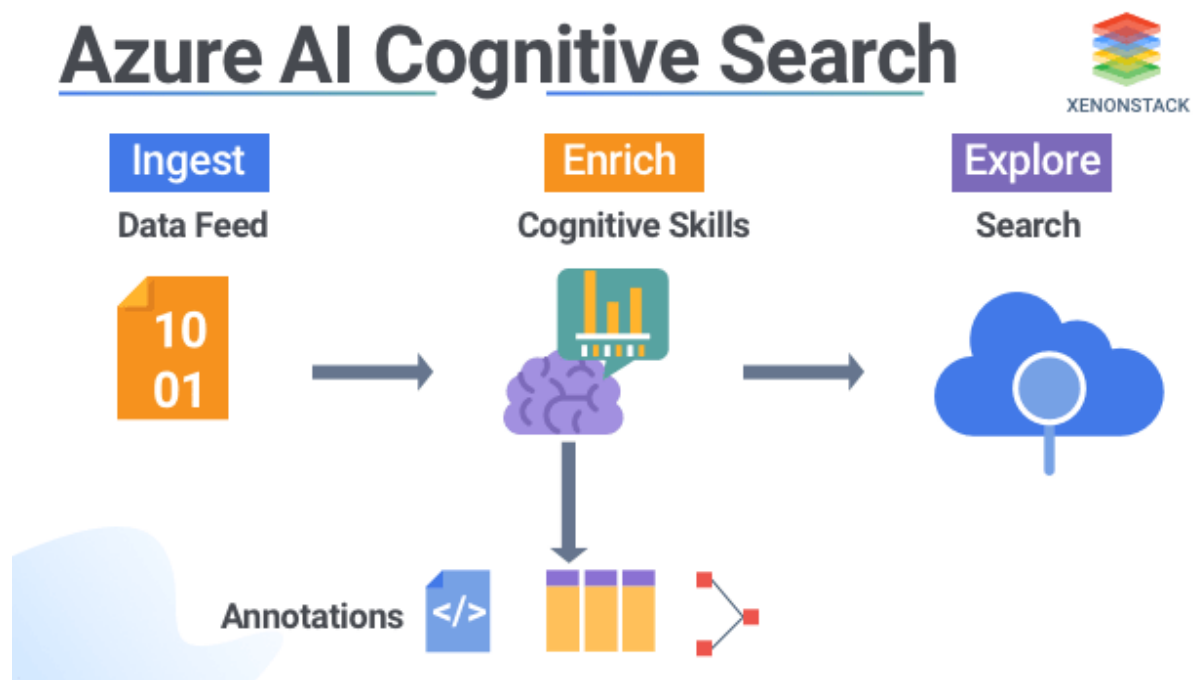
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App Url

<https://st10389916-hgb0fcfr6ccbccc.canadacentral-01.azurewebsites.net/>

Database Design and Cognitive Search

Azure Cognitive Search Compared to Traditional Search Engines



Introduction

Azure Cognitive Search is a Microsoft cloud-based search platform that incorporates artificial intelligence (AI) into the processes of indexing and retrieving data. In contrast to traditional search engines that depend mainly on keyword matching, Azure Cognitive Search uses technologies such as machine learning, natural language processing (NLP), and semantic search to deliver more accurate and context-aware results. This allows organizations to efficiently search and analyze both structured and unstructured information (bloomcs, 2025).

Traditional search engines are mainly built to retrieve information based on exact keywords or phrases entered by users. While this method is effective for basic searches, it often has difficulty understanding meaning, intent, and relationships between different pieces of data. Azure Cognitive Search enhances this capability by using AI-powered features that can process documents, images, scanned files, and multimedia content more intelligently (herrmannnglobal, 2024).

Differences Between Azure Cognitive Search and Traditional Search Engines

A key difference between Azure Cognitive Search and traditional search engines is how data is processed and interpreted. Traditional search engines mainly rely on keyword indexing and ranking methods to locate relevant information. Azure Cognitive Search, however, enhances this process through AI-powered features such as Optical Character Recognition (OCR), language detection, entity extraction, and sentiment analysis, which improve the accuracy and relevance of search results (herrmannnglobal, 2025).

Another significant distinction is scalability. Conventional search systems often require manual maintenance, infrastructure configuration, and hardware upgrades as workloads increase. Azure Cognitive Search is designed for cloud environments and can automatically scale based on organizational demands. This makes it highly effective for businesses that process and manage large volumes of data within distributed cloud systems (researchgate, 2024).

Feature	Traditional Search Engines	Azure Cognitive Search
Search Method	Keyword matching	AI-powered semantic search
Data Support	Mostly text-based data	Structured and unstructured data
AI Features	Limited	OCR, NLP, entity recognition
Scalability	Manual scaling	Automatic cloud scaling
Context Understanding	Basic	Advanced semantic understanding

Use Cases of Azure Cognitive Search

Azure Cognitive Search is especially useful in industries that manage large amounts of complex and unstructured data. In healthcare environments, the service can analyze patient records, prescriptions, laboratory results, and medical histories. This allows healthcare professionals to retrieve information through natural language searches instead of depending solely on exact keyword matches (preprints, 2023).

Within e-commerce platforms, Cognitive Search enhances product discovery and improves the customer experience by understanding the intent behind user searches. For instance, a user searching for “budget gaming laptop for students” can receive more relevant product recommendations based on context rather than simple keyword matching (preprints, 2023).

Legal firms and large enterprises also gain advantages from Cognitive Search because it can organize and analyze extensive collections of contracts, policies, and legal documents. Furthermore, media organizations use the technology to search multimedia content, subtitles, images, and video-related information more effectively (techtic, 2022).

Limitations and Mitigation Strategies

Despite its advantages, Azure Cognitive Search has several limitations. One major challenge is cost. AI enrichment features such as OCR and natural language processing require additional computing resources, which may increase operational expenses when processing very large datasets (infoservices, 2022).

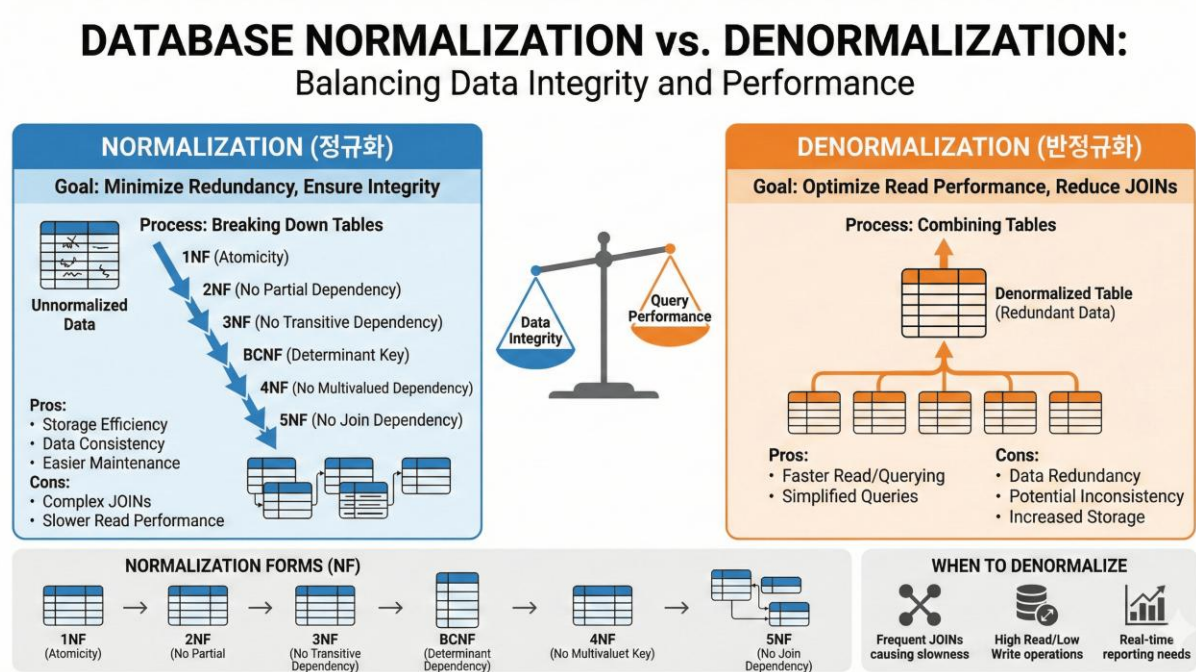
Another limitation is implementation complexity. Organizations must carefully design indexes, schemas, and search configurations to achieve accurate results. Poorly designed indexes may negatively affect search relevance and system performance (kpinfo.tech, 2025).

Conclusion

Azure Cognitive Search provides significant improvements over traditional search engines by introducing AI-driven, scalable, and context-aware search capabilities. Unlike conventional search systems that mainly rely on keyword retrieval, Azure Cognitive Search can analyze both structured and unstructured data intelligently. Although challenges such

as cost, implementation complexity, and AI inaccuracies exist, these issues can be minimized through proper design, optimization, and monitoring.

Importance of Database Normalization in Cloud-Based Database Design



Introduction

Database normalization is the process of organizing data into structured tables to reduce redundancy and improve consistency. In cloud environments such as Microsoft Azure, normalization plays an important role in improving database efficiency, maintainability, scalability, and data integrity (dbvis, 2023).

Cloud-based systems manage large amounts of distributed data, making efficient database design essential for maintaining high performance and reducing operational costs. Proper normalization ensures that resources are used effectively while keeping information accurate and consistent (dasca, 2022).

Importance of Database Normalization

One of the main purposes of normalization is to reduce duplicate data. By storing information only once and linking related tables together, organizations can minimize storage requirements and improve data organization. This is particularly important in cloud environments where storage and processing costs directly affect operational expenses.

Normalization also improves data integrity by ensuring that updates occur consistently across the database. Without normalization, duplicate records can create inconsistencies when data is updated in one location but not another. Proper normalization helps prevent insert, update, and delete anomalies (valoremreply, 2020).

Normalized vs Denormalized Structures

A normalized database structure separates data into multiple related tables to minimize redundancy and improve consistency. This design is commonly used in transactional systems where reliable updates and data accuracy are important.

A denormalized structure intentionally duplicates data to improve read performance and reduce complex joins. This approach is commonly used in reporting systems, analytics platforms, and distributed cloud databases where fast data retrieval is prioritized (splunk, 2024).

Feature	Normalized Databases	Denormalized Databases
Redundancy	Low	High
Storage Usage	Efficient	Increased
Query Performance	Slower joins	Faster reads
Data Integrity	Strong	Moderate
Best Use Case	Transaction systems	Analytics and reporting

Performance and Scalability in Azure

Normalized databases improve transactional performance because they maintain consistency and reduce redundant updates. In Azure SQL Database environments,

normalized structures are suitable for Online Transaction Processing (OLTP) systems such as healthcare, banking, and enterprise management applications (fabiantechtips, 2025).

However, normalized databases often require multiple table joins to retrieve information. In distributed cloud systems, these joins may increase latency and reduce performance when handling very large datasets.

Denormalized databases improve read performance because fewer joins are required. This makes them more appropriate for reporting systems, dashboards, and analytics platforms. In Azure Cosmos DB, denormalization is often preferred because embedding related information into documents reduces cross-partition queries and improves response times (exasol, 2023).

Conclusion

Database normalization is an important aspect of cloud-based database design because it improves consistency, maintainability, storage efficiency, and data integrity. In Azure environments, normalized databases are effective for systems that require reliable transactional processing. On the other hand, denormalized structures provide advantages for applications that require fast data retrieval and scalability. As a result, many modern cloud architectures combine both approaches to balance performance, scalability, and consistency effectively.

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